

Roll No. _____

Total No. of Pages: 02

Integrated M.C.A. (Semester - VII)
Examination - Dec. - 2025
IMCA - 703T
Theory of Computation

Time Allowed: Three Hours

Maximum Marks: 70
 Min. Marks :28

This question paper contains two sections as under :

Section-A**Max. Marks-20**

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×02=20)

Section-B**Max. Marks-50**

This section will have 10 questions, having 02 questions from each unit, one question to be attempt from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×05=50)

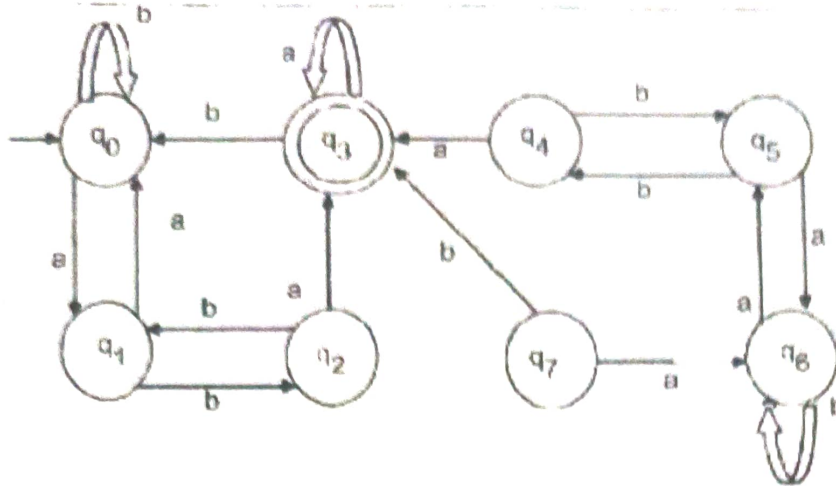
Section - A

- Q.1. Define the non deterministic finite Automata. ✓
- Q.2. Define the strings. ✓
- Q.3. Define the Regular set. ✓
- Q.4. Define Grammar. ✓
- Q.5. Define Parse tree. ✓
- Q.6. What is null production ? ✓
- Q.7. What is unit production ? ✓
- Q.8. What is halting problem ? ✓
- Q.9. What is time Complexity ? ✓
- Q.10. What is undecidability ? ✓

Unit - I

Q.2

Construct the minimum state automaton equivalent to the given diagram.



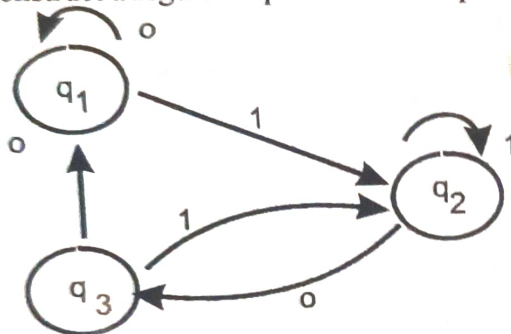
OR

Explain application of DFA in real life.

Unit - II

Q.3

Construct a regular expression corresponding to the given state diagram.



OR

Discuss the pumping lemma for regular languages with an example.

Unit - III

Q.4

(i)

Define :

Ambiguous grammar (ii) Derivation tree (iii) Chomsky Normal form

OR

Find a grammar in GNF equivalent to the grammar.

$E \rightarrow E+T \mid T, T \rightarrow T * F \mid F, F \rightarrow E \mid a$

Unit - IV

Q.5

Discuss closure properties of CFLs.

OR

Discuss :

(a) Deterministic pushdown automata (b) Non-deterministic pushdown automata

Unit - V

Q.6

Discuss variants of Turing machines :

OR

Define :

(i) Undecidability (ii) Halting problem (iii) Time complexity

834